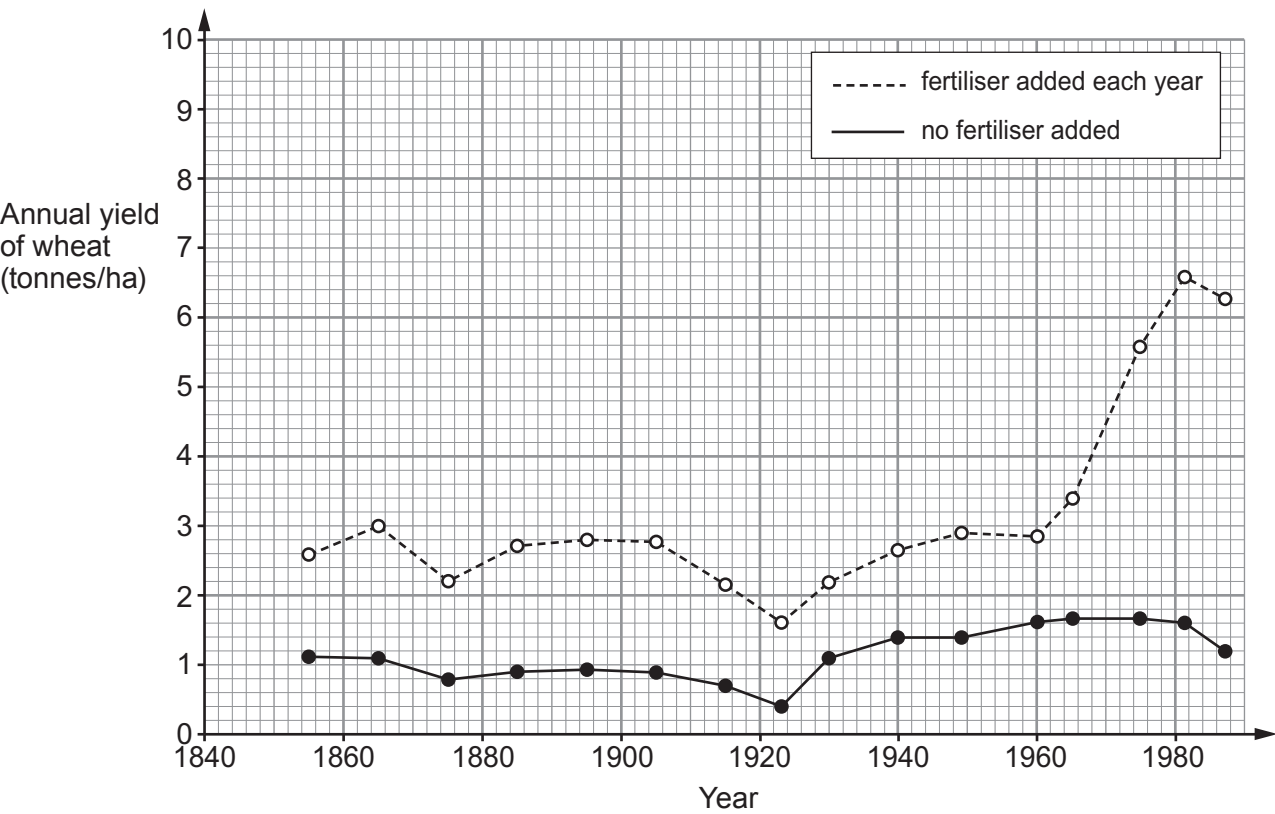


5. The information below is about the growth of wheat plants in a long-term investigation.



Wheat growth experiment at a research centre in Mid Wales

The graph shows the results of an investigation into the use of fertiliser. One field had the same mass of fertiliser added each year from 1855. Another field had no fertiliser added in that time.



Examiner
only

- (a)

(i)

From the graph, what evidence is there that:

I.

the addition of fertiliser increased the growth of wheat plants;

[1]

.....

II.

the fertiliser took a long time to reach its full effect?

[1]

.....

(ii)

How were the data from the field without fertiliser important to the scientists in interpreting their results?

[1]

.....
- (b)

Farmers now frequently add NPK fertiliser to their fields.
State the way in which the following mineral nutrients in NPK fertiliser affect the growth of their crop plants.

[2]

(i)

Effect of phosphate

(ii)

Effect of potassium

5

